

Nuclear sensors

Alpha, beta and gamma rays, as well as neutrons are the most common forms of ionising radiation. Workers in nuclear power plants receive low doses of radiation. Noting that exposure to high levels of ionising radiation can cause cancer, Atomic Energy Regulatory Board (AERB) prescribed the radiation dose limit for workers. Nuclear Power Corporation of India Limited (NPCIL) has strict procedures in place to keep radiation doses to workers within limits. Therefore radiation doses received by workers are recorded and monitored regularly.

Recent nuclear disaster in Japan has raised awareness about risks from nuclear radiation and created an additional demand for commercial as well as personal radiation detection devices.

Sensors and detectors play a major role in ensuring safety of nuclear plant workers. Nuclear particle sensors have been already developed to detect radiation types. Instead of tubes and chambers for nuclear particle detection, a variety of solidstate sensors are available today. Solidstate sensors directly convert the incident radiation into electrical current using materials such as silicon, germanium and cadmium zinc telluride. These sensors have high energy resolutions, which makes them suitable for detecting the exact amount of radiation energy. Alpha, beta, gamma and X-ray radiations can be detected using silicon PIN photodiodes.

Generally, solidstate radiation detectors can be divided into scintillator-based and direct-absorption-based detectors. Scintillator-based detectors provide better radiation detection but these are costlier. Direct-absorption detectors based on silicon are a reliable and inexpensive alternative. For example, First Sensor AG has developed a PIN diode based X100-7 direct-absorption detector for nuclear applications. Ultra Electronics, NSPI, USA, also designs and manufactures sensors for safety-related and critical measurements in nuclear power plants.

Autonomous vehicle sensors

Safety, fast identification, timely action, reliable notification and warning messages are of prime importance in autonomous vehicles. These are moni-

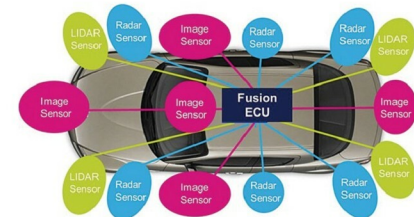


Fig. 6: Autonomous car sensors (Courtesy: www.sensorsmag.com)



Fig. 7: BGT24MTR11, a 24GHz chip for radar front-end application

tored and controlled by embedded MEMS sensors used in autonomous vehicles. Embedded sensor systems include cameras, radars, light-based radar (LiDAR) systems, ultrasonic sensors, wheel-speed sensors and global positioning systems (GPS).

Camera. Cameras and radar systems are prerequisites for all levels of automation. Using image sensors, cameras provide information like speed, distance and outlines of obstacles and moving objects. Autonomous cars use rear, front and 360-degree cameras. Cameras help the driver with a better representation of the surrounding and environment outside the vehicle.

Normally, four to six cameras are required to obtain realistic 3D images. Front camera systems are used to automatically detect objects within 100-250 metres. The algorithm in cameras can identify pedestrians, motor vehicles, side strips, road margins, etc. These can also detect traffic signs and signals. A good dynamic range of the camera is required to provide clear image even when sun rays are falling

directly onto the camera lens.

Radar. It is used for detection and localisation of objects using radio waves. Advanced driver-assistance systems employ many radar sensors making an important contribution to the overall function of autonomous driving. Many radar sensors and modules are already available in the market. For example, STRADA770 single-chip transceiver from STMicroelectronics is available in FO-WLP 230 package for automotive radar applications. It operates in the 76-81GHz frequency band. Another popular chip is Infineon's 24GHz BGT24MTR11 for radar front-ends.

LiDAR. It is a laser-based system primarily used to measure distances to stationary as well as moving objects. It calculates the distance of an object from the moving car. It employs special procedures to provide 3D images of the detected objects. The system is a complex mechanical mirror system, normally mounted on top of the autonomous car, which captures spatial images of objects with 360° all-round visibility.

Wheel-speed sensor. It is used to determine the speed at which the car is moving. Usually, it consists of a toothed ring and magnetic pickup used by anti-lock brake systems, traction control and other systems to indicate wheel speed.

Ultrasonic sensor. These are used for parking and detecting objects very close to the vehicle.

GPS. It provides the information for accurate positioning and location of the car. **EFY**